

Math 142, ~~519~~
Exam 1

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	21	
2	20	
3	22	
4	20	
5	22	
Total	105	

Good Luck !

Problem 1. (21 points) Evaluate the following integrals.

(a) $\int \cos(x) \sqrt{1 + \sin(x)} dx$

Let $u = 1 + \sin x \rightarrow du = \cos x dx$

$$\int \cos x \sqrt{1 + \sin x} dx = \int \sqrt{u} du$$

$$= \frac{2}{3} u^{\frac{3}{2}} + C = \frac{2}{3} (1 + \sin x)^{\frac{3}{2}} + C$$

(b) $\int \frac{x}{x^4 + 1} dx$

let $u = x^2 \rightarrow du = 2x dx$

$$\int \frac{x}{x^4 + 1} dx = \int \frac{1}{u^2 + 1} \frac{du}{2}$$

$$= \frac{1}{2} \arctan(u) + C = \frac{1}{2} \arctan(x^2) + C$$

(c) $\int_0^{\pi/4} \tan^2 \theta \sec^4 \theta d\theta$

Let $u = \tan \theta \rightarrow du = \sec^2 \theta d\theta$

When $\theta = 0$, $u = \tan 0 = 0$; $\theta = \frac{\pi}{4}$, $u = \tan(\frac{\pi}{4}) = 1$

$$\int_0^{\pi/4} \tan^2 \theta \sec^4 \theta d\theta = \int_0^{\pi/4} \tan^2 \theta \sec^2 \theta \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \tan^2 \theta (1 + \tan^2 \theta) \sec^2 \theta d\theta$$

$$= \int_0^1 u^2 (1 + u^2) du = \int_0^1 u^2 + u^4 du$$

$$= \left(\frac{u^3}{3} + \frac{u^5}{5} \right) \Big|_0^1 = \frac{1}{3} + \frac{1}{5} = \frac{8}{15}$$

Problem 2. (20 points) Evaluate the following integrals.

(a) $\int_0^1 x e^{-2x} dx$

Let $u = x$, $dv = e^{-2x} dx$
 $\downarrow \quad \quad \downarrow$
 $du = dx$, $v = -\frac{1}{2} e^{-2x}$

$$\begin{aligned}\Rightarrow \int_0^1 x e^{-2x} dx &= x \left(-\frac{1}{2} e^{-2x}\right) \Big|_{x=0}^{x=1} - \int_0^1 \left(-\frac{1}{2}\right) e^{-2x} dx \\ &= -\frac{1}{2} e^{-2} - \left(\frac{1}{4} e^{-2x}\right) \Big|_{x=0}^{x=1} \\ &= -\frac{1}{2} e^{-2} - \frac{1}{4} e^{-2} + \frac{1}{4} \\ &= -\frac{3}{4} e^{-2} + \frac{1}{4}\end{aligned}$$

(b) $\int \sin \sqrt{t} dt$

(Hint: first use the substitution $x = \sqrt{t}$)

Let $x = \sqrt{t}$, then $x^2 = t$, $2x dx = dt$

$$\int \sin \sqrt{t} dt = \int (\sin x)(2x) dx$$

$$= 2 \int x \sin x dx \leftarrow \begin{array}{l} \text{let } u = x, dv = \sin x dx \\ du = dx, v = -\cos x \end{array}$$

$$= 2 \left[-x \cos x - \int (-\cos x) dx \right]$$

$$= -2x \cos x + 2 \int \cos x dx$$

$$= -2x \cos x + 2 \sin x + C$$

$$= -2\sqrt{t} \cos \sqrt{t} + 2 \sin \sqrt{t} + C$$

Problem 3. (22 points) Evaluate the following integrals.

(a) $\int \sqrt{4-t^2} dt$

Let $t = 2\sin\theta$, then $dt = 2\cos\theta d\theta$ and $\sqrt{4-t^2} = 2\cos\theta$.

$\theta = \arcsin\left(\frac{t}{2}\right)$

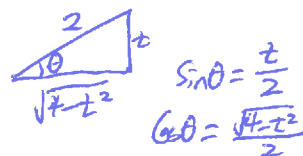
$$\int \sqrt{4-t^2} dt = \int 2\cos\theta \cdot 2\cos\theta d\theta = \int 4\cos^2\theta d\theta = \int 4 \cdot \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= \int (2+2\cos 2\theta) d\theta = 2\theta + \sin 2\theta + C$$

$$= 2\arcsin\left(\frac{t}{2}\right) + 2\sin\theta\cos\theta + C$$

$$= 2\arcsin\left(\frac{t}{2}\right) + 2\left(\frac{t}{2}\right) \cdot \left(\frac{\sqrt{4-t^2}}{2}\right) + C$$

$$= 2\arcsin\left(\frac{t}{2}\right) + \frac{t\sqrt{4-t^2}}{2} + C$$



(b) $\int \frac{2x+4}{x^3-x} dx$

$$x^3-x = x(x^2-1) = x(x-1)(x+1)$$

$$\frac{2x+4}{x^3-x} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$$

$$= \frac{A(x-1)(x+1) + Bx(x+1) + Cx(x-1)}{x(x-1)(x+1)} = \frac{A(x^2-1) + B(x^2+x) + C(x^2-x)}{x(x-1)(x+1)}$$

$$= \frac{(A+B+C)x^2 + (B-C)x - A}{x(x-1)(x+1)}$$

$$\Rightarrow \begin{cases} A+B+C=0 \\ B-C=2 \\ -A=4 \end{cases} \Rightarrow \begin{cases} A=-4 \\ B=3 \\ C=1 \end{cases}$$

Thus $\int \frac{2x+4}{x^3-x} dx = \int \frac{-4}{x} + \frac{3}{x-1} + \frac{1}{x+1} dx$

$$= -4\ln|x| + 3\ln|x-1| + \ln|x+1| + C$$

Problem 4. (20 points) Evaluate each of the following improper integrals or show that it is divergent.

(a) $\int_2^{+\infty} \frac{dx}{x(\ln x)^2}$

$$= \lim_{b \rightarrow +\infty} \int_2^b \frac{dx}{x(\ln x)^2}$$

$$= \lim_{b \rightarrow +\infty} \left[-\frac{1}{\ln x} \right]_{x=2}^{x=b}$$

$$= \lim_{b \rightarrow +\infty} -\frac{1}{\ln b} + \frac{1}{\ln 2} = 0 + \frac{1}{\ln 2} = \frac{1}{\ln 2}$$

$\Rightarrow$ Convergent

(b) $\int_2^{10} \frac{dy}{\sqrt[3]{10-y}}$

$$= \lim_{b \rightarrow 10^-} \int_2^b \frac{dy}{\sqrt[3]{10-y}} = \lim_{b \rightarrow 10^-} \int_2^b (10-y)^{-\frac{1}{3}} dy$$

$$= \lim_{b \rightarrow 10^-} \left[-\frac{3}{2} (10-y)^{\frac{2}{3}} \right]_{y=2}^{y=b}$$

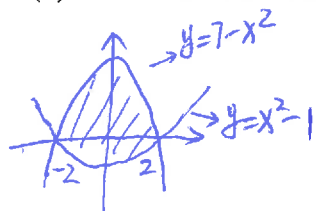
$$= \lim_{b \rightarrow 10^-} -\frac{3}{2} (10-b)^{\frac{2}{3}} + \frac{3}{2} (8)^{\frac{2}{3}}$$

$$= 0 + \frac{3}{2} \cdot 4 = 6$$

$\Rightarrow$ Convergent

Problem 5. (22 points)

(a) Find the area of the region enclosed by the curves $y = 7 - x^2$ and $y = x^2 - 1$.



$$\begin{cases} y = 7 - x^2 \\ y = x^2 - 1 \end{cases} \rightarrow \begin{aligned} 7 - x^2 &= x^2 - 1 \\ 2x^2 &= 8 \Rightarrow x^2 = 4 \Rightarrow x = 2, -2 \end{aligned}$$

$$A = \int_{-2}^2 (7 - x^2) - (x^2 - 1) dx$$

$$= \int_{-2}^2 8 - 2x^2 dx$$

$$= \left[8x - \frac{2}{3}x^3 \right]_{-2}^2$$

$$= \left(16 - \frac{16}{3} \right) - \left(-16 + \frac{16}{3} \right)$$

$$= \frac{32}{3} + \frac{32}{3} = \frac{64}{3}$$

(b) Use the trapezoidal rule and Simpson's rules respectively with $n = 6$ to estimate

$$\int_{-1/2}^{1/2} \cos^2(\pi x) dx.$$

$$a = -\frac{1}{2}, b = \frac{1}{2}, n = 6$$

$$\Delta x = \frac{\frac{1}{2} - (-\frac{1}{2})}{6} = \frac{1}{6}$$

i	x_i	$y_i = f(x_i)$
0	$-\frac{1}{2}$	0
1	$-\frac{1}{3}$	$\frac{1}{4}$
2	$-\frac{1}{6}$	$\frac{3}{4}$
3	0	1
4	$\frac{1}{6}$	$\frac{3}{4}$
5	$\frac{1}{3}$	$\frac{1}{4}$
6	$\frac{1}{2}$	0

$$f(x) = \cos^2(\pi x)$$

Trapezoidal Rule:

$$T_4 = \frac{\Delta x}{2} (y_0 + 2y_1 + 2y_2 + 2y_3 + 2y_4 + 2y_5 + y_6)$$

$$= \frac{1}{6} \cdot \frac{1}{2} (0 + 2 \cdot (\frac{1}{4}) + 2(\frac{3}{4}) + 2(1) + 2(\frac{3}{4}) + 2(\frac{1}{4}) + 0)$$

$$= \frac{1}{12} (6) = \frac{1}{2}$$

Simpson's rule:

$$S_4 = \frac{\Delta x}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + 4y_5 + y_6)$$

$$= \frac{1}{6} \times \frac{1}{3} (0 + 4 \cdot (\frac{1}{4}) + 2(\frac{3}{4}) + 4(1) + 2(\frac{3}{4}) + 4(\frac{1}{4}) + 0)$$

$$= \frac{1}{18} (9) = \frac{1}{2}$$